

Four-Fold Cross-Validation: PINN-LSTM vs. LSTM Baseline

IGBT RUL Reproduction of Lu et al. (2023), NASA PCoE Dataset #8

Leave-one-device-out | MSE $\times 10^{-3}$ | lower MSE / higher R^2 = better | $\Delta\text{MSE} > 0$ and $\Delta\text{MSE}\% > 0$ mean PINN beat baseline

Test device	LSTM MSE	LSTM R^2	PINN-LSTM MSE	PINN-LSTM R^2	ΔMSE (abs)	ΔMSE (%)	Lu PINN MSE	Lu PINN R^2	dev MSE	dev R^2
2	9.48	0.877	9.28	0.880	0.20	+2.1%	7.29	0.881	2.00	0.001
3	5.95	0.922	8.33	0.891	-2.38	-40.0%	1.65	0.978	6.68	0.087
4	17.06	0.778	19.99	0.740	-2.93	-17.2%	5.96	0.898	14.03	0.158
5	3.71	0.952	5.35	0.930	-1.64	-44.2%	4.09	0.954	1.27	0.024
AVG	9.05	0.882	10.74	0.860	-1.69	-18.6%	4.75	0.930	5.99	0.070

Caption. On this run the physics-informed LSTM did not improve on the vanilla LSTM baseline: averaged across the four folds its test MSE was 18.6% higher (worse) and mean R^2 fell from 0.882 to 0.860 — the opposite of the +13.9% MSE improvement Lu et al. (2023) report for PI-LSTM. The vanilla-LSTM baseline R^2 values are nonetheless reasonable (mean 0.882), confirming the recurrent model fits the degradation signal once the paper architecture (80-cell single-layer LSTM, seq=10) is used. The largest deviation from Lu et al. is on the device-4 fold ($|\Delta\text{MSE}| = 14.0 \times 10^{-3}$, $|\Delta R^2| = 0.158$), where both models fit worst. [FILL IN after inspecting device 4: the most likely driver is the automatic end-of-life detection (FAILURE_DROP_THRESHOLD) and resulting RUL labels on this device's noisy late-life V_{ce} ; confirm whether the detected N_f matches the paper's hand-identified failure point, and note the average-downsampling window vs the paper's per-square-wave-cycle averaging.] All runs use the Lu-matching configuration (80-cell single-layer LSTM, sequence length 10, $\alpha=0.1$, $\beta=100$, $\gamma=0.1$, Adam, 2000 epochs) under leave-one-device-out cross-validation on NASA PCoE Dataset #8. The physics terms add little here because the monotonic-decreasing penalty sees only small violations under heavy window overlap, while the $\beta=100$ boundary penalty can distort training when predictions stray slightly outside $[0,1]$. Lu et al. PI-LSTM reference values are read from their Figure 10 ($\times 10^{-3}$).

Generated from outputs/four_fold_comparison.csv. Lu et al. (2023), Sci. Rep. 13:10167 (CC BY 4.0); reference values from Figure 10.